Dequantization on Geometry

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I'm interested in the re-interpretation of non-linear PDE from algebraic geometry and QFT perspective. It's typically Navier-Stokes, Apery, period integral of Landau Ginzburg, Monge-Ampere below, and I consider possible strategies are 1: I consider non-linear PDE differential geometrically, by making some connection form diffyq $\nabla f = 0$ on a real smooth manifold X, possibly symplectification (X', ω) if necessary, where I further put geometric quantization, and 2: Derived moduli space of solutions spaces of non-linear PDE, whose algebraic geometric realization, and 3: Transform non-linear PDE to CFT, corresponding to algebraic geometry.

- Navier Stokes

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

  Or

$$\partial_\mu T^{\mu\nu} = 0$$

- Apery (Quantum Satake)

$$\mathcal{L}_{\mathrm{Ap}} = \theta^3 - z(2\theta+1)(17\theta^2+17\theta+5) + z^2(\theta+1)^3, \qquad \theta = z\frac{d}{dz}.$$

- Landau-Ginzburg (Period Integral, SCFT)

$$\int e^{-f(x)} \qquad f(x) = \text{polynomial}$$

- Monge-Ampere (SYZ)

$$\det\left(\frac{\partial^2 \phi}{\partial x_i \partial x_j}\right) = f(x)$$

- SUGRA (String Theory)

$$R_{\mu\nu} - \frac{1}{2}R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Remark 0.1

I also noticed several different approaches, Kcryzka introduces derived algebraic geometry realization of jet spaces, Ladychenskaya formula for solving 2d NSE. These are quite interesting but quite different from my study.

keywords:
picard-fuch, quantum connection.

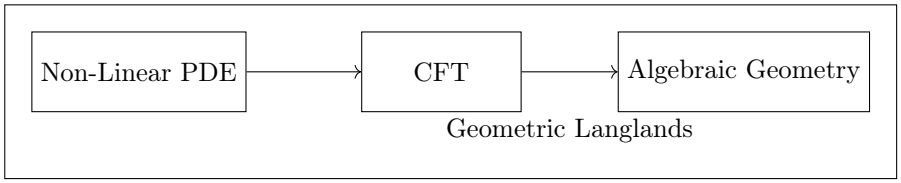
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1 Navier-Stokes



Proposition 1.1. NSE-CFT Conversion

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

transformed into

$$\partial_\mu T^{\mu\nu} = 0$$

The question.

Question 1.1

- Connection form diffyq
- Symplectic extension
- Quantization

All real smooth manifold can be a Lagrangian submanifold of some Symplectic manifold. In particular cotangent bundle, $\iota : M \hookrightarrow T^*M$. How does this embedding maps connection form diffyq $\nabla f = 0$? In addition, connection ∇ on M needs to have natural extension $\tilde{\nabla}$ on T^*M, so its restriction is $\tilde{\nabla}|_M = \nabla$.

$$\begin{aligned} M &\hookrightarrow T^*M \\ \nabla &\mapsto \tilde{\nabla} \end{aligned} \tag{1}$$

I define quantization of non-linear PDE by quantization of connection form ∇ in symplectic geometry.

Definition 1.1

Definition 1.2. Quantization of NSE

2 Apery

Definition 2.1. Geometric Satake

$$\mathrm{Perv}(\mathrm{Gr}_G) \simeq \mathrm{Rep}(G^\vee).$$

Definition 2.2. Quantum Satake

Definition 2.3. 4D/2D CFT Correspondence

Definition 2.4

Definition 2.5

Definition 2.6

3 Quantization

One of my interests in non-linear PDE is to build framework of quantization, which is first of all embedding of connection form $\nabla f = 0$ to symplectic geometry, and secondly to do quantization, possibly Fedosov's method of geometric construction by deformation quantization, or geometric prequantization: these two are different, and I don't know which suits better. In any case, quantization of Navier-Stokes equation is consider to be difficult according to ChatGPT.

Definition 3.1. Geometric Prequantization

Prequantization is a process of making an operator $Q(f)$ from a smooth function f, $Q : C^\infty(X) \to End(\mathcal{H})$ on a symplectic manifold (X, ω). More precisely, $\mathcal{H} = \Gamma^\infty(L)$ where L is a prequantum line bundle with a connection ∇_{X_f}, and an operator given by $Q(f) := -i\hbar\nabla_{X_f} + f$, so that bracket satisfies $[Q(f), Q(g)] = i\hbar Q(\{f, g\})$.

Typically, the operator is Kostant–Souriau operator $Q(f) := -i\hbar(X_f + \frac{1}{i\hbar}\theta(X_f)) + f$.

Definition 3.2. Deformation Quantization

Deformation quantization transforms ordinary product to star product $fg \mapsto f \star g$ where the star product is defined by

$$f \star g = \Sigma_{n\geqq 0}\frac{1}{n!}(\frac{i\hbar}{2})^n \Pi^n(f, g)$$

where Π is Poisson bivector, and $\Pi^1(f, g) = \{f, g\}$ is ordinary bracket. Naturally, the Poisson bracket is defined by

$$\frac{f \star g - g \star f}{i\hbar} = \{f, g\} + O(\hbar)$$

Fedosov's method of deformation quantization in symplectic geometry.

Definition 3.3. Fedosov's Deformation Quantization

TBA

Definition 3.4

References

[1] Luiz Carazolli et al. *A quantum connection approach to rationality*. 2026. arXiv: 2610.06669 [math.AG]. URL: https://arxiv.org/abs/2610.06669.

[2] Claudio Emmrich and Alan Weinstein. *The differential geometry of Fedosov's quantization*. 1993. arXiv: hep-th/9311094 [hep-th]. URL: https://arxiv.org/abs/hep-th/9311094.

[3] Roman Fedorov, Alexander Soibelman, and Yan Soibelman. *Motivic invariants of moduli stacks of Higgs bundles and bundles with connections: results and speculations*. 2026. arXiv: 2602.07713 [math.AG]. URL: https://arxiv.org/abs/2602.07713.

[4] Edward Frenkel. *Lectures on the Langlands Program and Conformal Field Theory*. 2005. arXiv: hep-th/0512172 [hep-th]. URL: https://arxiv.org/abs/hep-th/0512172.

[5] Jacob Kryczka and Artan Sheshmani. *The $\mathcal{D}$-Geometric Hilbert Scheme – Part I: Involutivity and Stability*. 2025. arXiv: 2507.07937 [math.AG]. URL: https://arxiv.org/abs/2507.07937.

[6] Daniel Pomerleano and Paul Seidel. *The quantum connection, Fourier-Laplace transform, and families of A-infinity-categories*. 2026. arXiv: 2308.13567 [math.SG]. URL: https://arxiv.org/abs/2308.13567.

[7] Sebastian Ali Sacasa-Cespedes. *A Geometric Approach to the Navier-Stokes Equations*. 2024. arXiv: 2411.18724 [physics.flu-dyn]. URL: https://arxiv.org/abs/2411.18724.

[8] Pavel Safronov. *Shifted geometric quantization*. 2020. arXiv: 2011.05730 [math.SG]. URL: https://arxiv.org/abs/2011.05730.

[9] Géry de Saxcé. *Symplectic and variational formulations of compressible and incompressible Navier-Stokes equation*. 2023. arXiv: 2306.04405 [math-ph]. URL: https://arxiv.org/abs/2306.04405.

[10] Yan Soibelman. *Moduli space of Conformal Field Theories and non-commutative Riemannian geometry*. 2025. arXiv: 2506.00896 [hep-th]. URL: https://arxiv.org/abs/2506.00896.

[11] Bertrand Toen. *Derived Algebraic Geometry and Deformation Quantization*. 2014. arXiv: 1403.6995 [math.AG]. URL: https://arxiv.org/abs/1403.6995.